CEVE 543 Final Project

Linear and Non-Linear Dimensionality Reduction in Climate Science: Methods and Applications of PCA and Autoencoders

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Introduction: What is Dimensionality Reduction?

The size and scale of high dimensional data is vast and noisy. This is particularly true in the field of climate science. Within the field of climate science, dimensionality reduction methods can be used to simplify massively complex datasets, identify major patterns of variability, and improve the accuracy and interpretability of climate models. This helps to differentiate between climate noise, external forcings, internal variability signals, and human-driven climate responses(DelSole and Tippett 2022).

Principal Component Analysis (PCA)

What is PCA?

PCA, or Principal Component Analysis, is a dimensionality reduction technique that reduces the dimensionality of data by transforming it into a new set of orthogonal axes (called principal components) that capture the most variance. This technique projects data onto these components in a way that maximizes the variance in fewer dimensions (DelSole and Tippett 2022).

How does PCA work?

First, standardize your data by normalizing (subtracting the mean and dividing by standard deviation for each feature to ensure that feature has equal weight for PCs). Then, compute the covariance matrix to identify how variables relate to each other. Calculate eigenvectors and eigenvalues of your covariance matrix to determine the direction and strength of variance in your data. Finally, project your data onto top principal components (DelSole and Tippett 2022).

Strengths, Limitations, and Appropriate Use Cases of PCA:

For small to moderately-sized, linearly separable datasets, PCA is an effective, computationally efficient method that provides an intuitive understanding of variance and how features relate to each other. Further, the eigenvalue decomposition and modern linear algebra libraries that serve as the mathematical foundation for this technique are well-studied. Disadvantages of this technique include PCA's sensitivity to scale, its reliance on variance, and the linear assumptions this technique makes. Because PCA is sensitive to the scale of input variables, variables with larger numerical ranges will dominate PCs. Additionally, PCA assumes that the directions of maximum variance are the most important feautures in a dataset, and this does not hold true in certain situations. Standardization can address this problem, but may change the interpretation of results. PCA also only captures linear relationships between variables. So, if your data has non-linear patterns, PCA may not capture the complexity of variable interactions. Thus, PCAs should be used when attempting to extract the most dominant features in high dimensional datasets where multi-colinearity exists between variables [Idrees (2024)](DelSole and Tippett 2022).

PCA Mathematical Framework (DelSole and Tippett 2022)

Key Assumptions:

This approach assumes that your data is discretized, a linear combination of orthogonal basis vectors, and mean centered.

Following DelSole & Tippett (2022)'s PCA framework, we can represent a spatiotemporal climate field as:

$$Y : \{1, \dots, N\} \times \{1, \dots, M\} \to \mathbb{R}$$

where (Y(t,s)) represents the observed value at time index (t) and spatial index (s). The matrix representation of the dataset is

$$Y \in \mathbb{R}^{N \times M},$$

with rows representing time and columns representing spatial locations.

$$Y(t,s) \approx \sum_{k=1}^{K} f_k(t)\, e_k(s).$$

In matrix form, the data is an (N ×M) matrix (Y),
with each row a time step and each column a spatial location.

Goal of PCA

Find a low-rank approximation of (Y) using (K) modes:

$$Y_K = \sum_{k=1}^{K} \mathbf{f}_k\, \mathbf{e}_k^T.$$

PCA Optimization Problem:

The first principal component solves:

$$\min_{\mathbf{f}_1,\,\mathbf{e}_1} \left\| Y - \mathbf{f}_1 \mathbf{e}_1^T \right\|_F^2,$$

After removing the first component, the PCA is applied to the residual:

$$Y - f_1 e_1^T.$$

Solution via Singular Value Decomposition (SVD):

PCA is equivalent to performing the SVD:

$$Y = USV^T,$$

where

- $U \in \mathbb{R}^{N\times R}$ contains temporal left singular vectors,
- $V \in \mathbb{R}^{M\times R}$ contains spatial right singular vectors (EOFs),
- $S = \mathrm{diag}(s_1, \dots, s_R)$ contains singular values,
- $R = \mathrm{rank}(Y)$.

The best rank-(K) approximation is:

$$Y_K = \sum_{k=1}^{K} s_k\, u_k v_k^T.$$

Identifying PCs and EOFs

PCs and EOFs can be defined as:

$$\mathbf{f}_k = s_k\, u_k, \qquad \mathbf{e}_k = v_k.$$

Because of this, the PCA reconstruction becomes:

$$Y_K = f_1 e_1^T + \cdots + f_K e_K^T.$$

Here,

- $\mathbf{e}_k$ are **Empirical Orthogonal Functions (EOFs)**,
- $\mathbf{f}_k$ are **Principal Components (PCs)**.

Variance Explained:

Because singular vectors are orthogonal:

$$\|Y\|_F^2 = \sum_{k=1}^{R} s_k^2.$$

The fraction of variance explained by the first (K) modes is:

$$\frac{\sum_{k=1}^{K} s_k^2}{\sum_{k=1}^{R} s_k^2}.$$

Example of PCA:

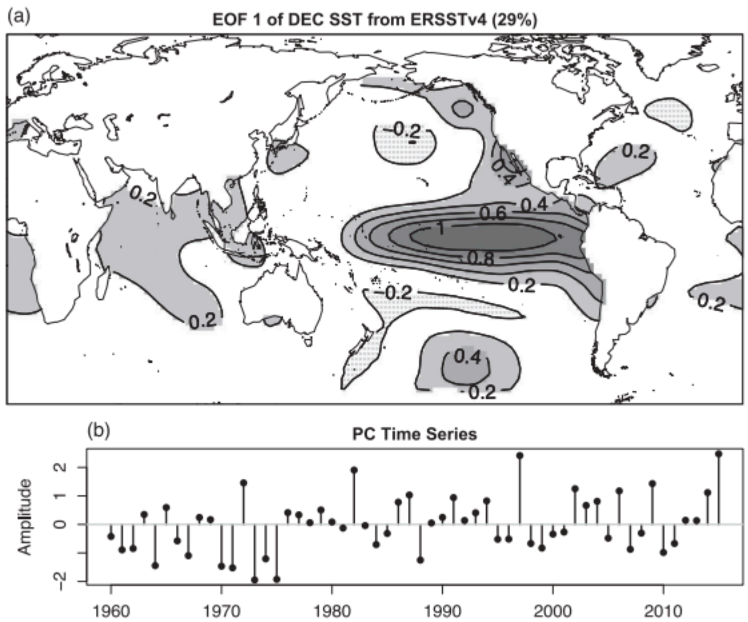

Figure 12.1 (a) Leading EOF of the December SST from the Extended Reconstructed SST of Huang et al. (2015). The percent variance explained by the leading component is indicated in the title. The value at each grid point gives the departure from climatology in units of degrees Kelvin per unit PC time series. (b) The corresponding leading PC time series.

Example of a PCA output (DelSole and Tippett 2022). Shows leading EOF explaining global December SSTs .

Autoencoders

What are Autoencoders?

Autoencoders are a type of unsupervised, neural network-based, dimensionality reduction technique that compresses high-dimensional, non-linear data into a lower-dimensional space, then reconstructs, or decodes, the original data from this compressed representation. Input and output dimensions are kept the same. [Michelucci (2022)](Gámez et al. 2004).

How do Autoencoders Work?

The first part of an autoencoder framework compresses input data into a lower dimensional latent space called a bottleneck. This aims to capture the initial data's most important features in fewer dimensions. The compressed verson of this data is represented in the bottleneck layer, and this has fewer neurons than the input. The decoder, or the second part of the autoencoder network, attempts to reconstruct the original input data from the compressed latent space (Michelucci 2022).

Strengths, Limitations, and Appropriate Use Cases of Autoencoders:

An advantage of autoencoders is their ability to reduce the dimensionality of complex, non-linear datasets (ex: images and text) where linear methods like PCA may fail. Additionally, autoencoders can accomodate a variety of different input data and can be used for a wide breadth of tasks, like feature extraction, data compression, image denoising, anomaly detection and facial recognition. Disadvantage-wise, autoencoders are computationally expensive, require a large amount of data to train effectively, and the features learned in the latent space (results of autoencoders) can be hard to interpret. Further, as with many deep learning techniques, autoencoders suffer from the black box problem. Use wise, autoencoders should be used when trying to identify and replicate meaningful patterns in very large, non-linear datasets with abstract bounds (ex: images, pictures, sound clips, etc.) [Gámez et al. (2004)](Idrees 2024) [Myrzaliyeva (n.d.)](Saenz, Lubbers, and Urban 2016).

Autoencoder Mathematical Framework (Michelucci 2022):

Key Assumptions:

Because autoencoders are unsupervised, the training datasets for autoencoders are assumed to be unlabeled. A training dataset for autoencoder use can be expressed as:

$$S_T = \{\mathbf{x}_i \mid i = 1, \ldots, M\}.$$

Where each $\mathbf{x}_i$ represents each input training observation and M represents all our observations.

Encoder:

In standard autoencoder architectures, the encoder and the decoder are neural networks.

The encoder takes the original high dimensional data and compresses it into a smaller, low-dimensional vector (hi); this vecotr is called the latent feauture representation, or the bottlenec. In general, the encoder can be written as a function g that will depend on some parameters.

$$g : \mathbb{R}^n \to \mathbb{R}^q, \qquad \mathbf{h}_i = g(\mathbf{x}_i).$$

Latent Feauture Representation/Bottleneck:

The latent representation is a low-dimensional, compressed representation of the original input data that captures the most meaningful feautures of the input data.

$$\mathbf{h}_i$$

Decoder:

The decoder reconstructs the input from the latent representation or bottleneck:

$$f : \mathbb{R}^q \to \mathbb{R}^n, \qquad \tilde{\mathbf{x}}_i = f(\mathbf{h}_i) = f(g(\mathbf{x}_i)).$$

Reconstruction Loss:

The autoencoder is trained by minimizing reconstruction error, or the difference between the input data and the reconstructed input. This can be represented by the following function:

$$\arg\min_{f,g} \frac{1}{M} \sum_{i=1}^{M} \Delta(\mathbf{x}_i,\, f(g(\mathbf{x}_i)))$$

Loss functions quantify how different the reconstructed output is from the input data. Two widely used loss functions include the Mean Square Error loss function (typically used for continuous data) and the Binary Cross Entropy loss function (typically used for input data that is normalized/scaled to [0,1]).

Common Loss Functions:

Mean Squared Error (MSE):

$$L_{\text{MSE}} = \text{MSE} = \frac{1}{M}\sum_{i=1}^{M}\|\mathbf{x}_i - \tilde{\mathbf{x}}_i\|^2 \tag{9}$$

Binary Cross-Entropy (BCE):

$$L_{\text{CE}} = -\frac{1}{M}\sum_{i=1}^{M}\sum_{j=1}^{n}\left[x_{j,i}\log\tilde{x}_{j,i} + (1 - x_{j,i})\log(1 - \tilde{x}_{j,i})\right]$$

A Typical Autoencoder Framework:

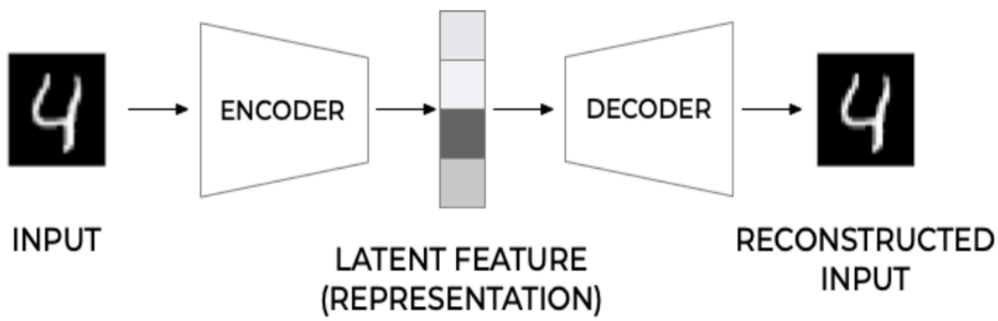

Figure 1: General structure of an autoencoder.

Typical Structure of an Autoencoder (Michelucci 2022).

Example of an Autoencoder:

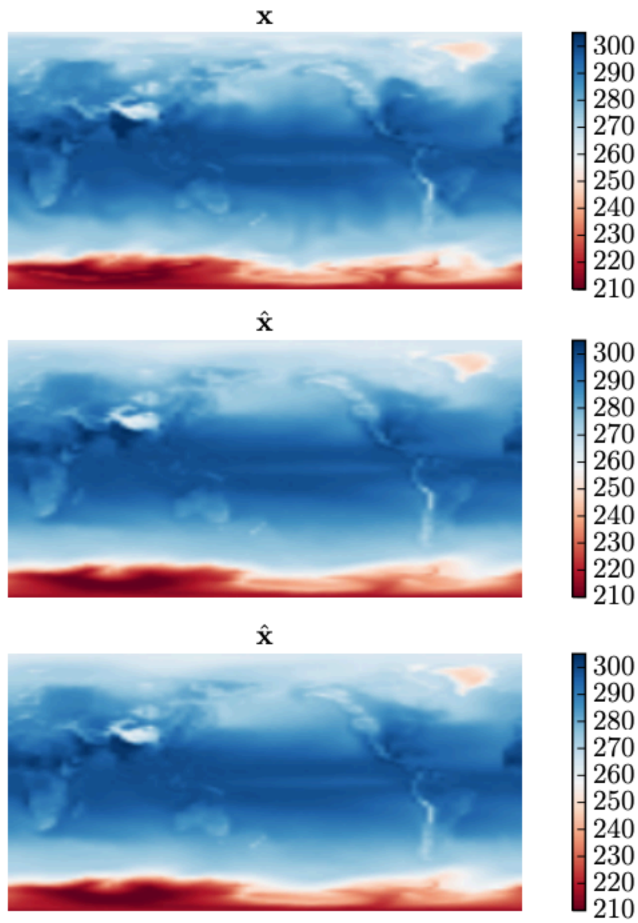

Fig. 2. Comparison of T_{as} [°K] from the original dataset (top) and the reconstruction using architectures B1 (middle) and B2 (bottom), sample 28618 from the IPSL-CM5A-LR dataset.

Example of an autoencoder output within context of surrogate climate modeling. In this figure, reasearchers are attempting to reproduce 150 years of pre-industrial mean SST anomalies from the IPSL climate model. Latent space outputs from two autoencoder architectures, B1 and B2, are being compared to initial input data(Saenz, Lubbers, and Urban 2016).

Application of Autoencoders to Class:

CEVE 543's coursework centers around quantifying hydroclimate extremes, their uncertainties, and the related impacts on relevant communities using varying statistical methods. By capturing low-dimensional, meaningful feautures from noisy, high-dimensional data, autoencoders and PCA can aid in the attribution and forecasting of hydroclimate extremes. For example, autoencoders can be be used to create informed lake discharge projections, an especially important task under the growing threat of increased climate variability and intensive land use/land changes. Autoencoders and PCA can also identify underlying linear and nonlinear climate patterns (ex: modes of internal variability like AMO and ENSO) driving pluvials, droughts, and seasonal rainfall patterns, enhancing the creation of tools like regional downscaling models and surrogate climate models. Additionally, dimensionality reduction underpins the computational framework of rare precipitation event detection, even without prior data (Saenz, Lubbers, and Urban 2016) (Murakami et al. 2022).

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